



DBMS 02 | Key Concepts & Finding Number of Candidate keys | CS & IT | GATE 2024 FastTrack Batch

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Summary

AI Notes

AI Slides

Text Notes

Screenshots

✦ 0

✦ 1

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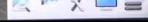
📷 13

**Topic: Attribute closure $[X]^+$** 

Example:
Let us consider a relation with attributes A, B, C, D, E, and F. Suppose that this relation has the FD's $AB \rightarrow C$, $BC \rightarrow AD$, $D \rightarrow E$, and $CF \rightarrow B$.
What is the closure of $\{A, B\}$, that is, $\{A, B\}^+$?

FD's : $\{AB \rightarrow C, BC \rightarrow AD, D \rightarrow E, CF \rightarrow B\}$

$\{AB\}^+ = \{ABCDE\}$



▶ 9:49

[MCQ]

[GATE:2 Marks]



#Q. The following functional dependencies are given:

[ISRO]

$AB \rightarrow CD$, $AF \rightarrow D$, $DE \rightarrow F$, $C \rightarrow G$, $F \rightarrow E$, $G \rightarrow A$.
Which one of the following options is false?

- ☒ A True $\{CF\}^+ = \{ACDEFG\}$ $\{CF\}^+ = [CFGEAD] - [ACDEFG]$
- ☐ B $\{BG\}^+ = \{ABCDG\}$ $\{BG\}^+ = [BGACD] \Rightarrow [ABCDG]$
- ☒ C False $\{AF\}^+ = \{ACDEFG\}$ $\{AF\}^+ = [AFED]$
- ☒ D False $\{AB\}^+ = \{ABCDG\}$ $\{AB\}^+ = [ABCDG]$

▶ 24:13

Topic: keys Concept

$R(ABCDE)$ $[A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E]$

$\{A\}^+ = \{ABCDE\}$

$\{B\}^+ = \{BCDE\}$

$\{C\}^+ = \{CDE\}$

$\{D\}^+ = \{DE\}$

$\{E\}^+ = \{E\}$

A is super key.

A is super key.
Any Super Set of super key A
is also super key.

$\begin{matrix} \underline{A} \\ \underline{AB} \\ \underline{AC} \\ \underline{ABC} \\ \vdots \\ \underline{ABCDE} \end{matrix}$ } all are super keys.

▶ 46:56

Topic: keys Concept

SUPER KEY : Let R be the Relational Schema
 X be Attribute set of Relation R .

If $(X)^+$ determines all attributes of Relational Schema R then X is Super key

OR

All attribute of Relation R is determined by attribute set of X then X is Super key.

▶ 47:05

Topic: keys Concept

Candidate key :

Minimal of Super key.

OR

Any Proper subset of Super key is also Super key then that Proper subset is Called Candidate key (2.50 on)

▶ 54:49

$R(ABCDE) [AB \rightarrow C, C \rightarrow D, B \rightarrow EA]$

Solⁿ $(AB)^+ = [ABCDE]$

AB is Super key.

AB
 Proper Subset
 $A \quad B$

$(A)^+ = [A]$
 $(B)^+ = [BEACD]$

B is Candidate key.

$R(ABCDE) [AB \rightarrow C, C \rightarrow D, B \rightarrow E]$

Solⁿ $(AB)^+ = [ABCDE]$

AB is Super key.

AB
 Proper Subset
 $A \quad B$

$(A)^+ = [A]$
 $(B)^+ = [BE]$

AB is Candidate key.

▶ 1:00:44



Topic: keys Concept



Candidate key :

Minimal of Super key.

(OR)

If Any Proper subset of Super key is also Super key
then that Proper subset is Called Candidate key (2.50 on)

▶ 1:06:03



key: Definitions



Superkey

- a set of attributes whose values together uniquely identify a tuple in a relation

Candidate Key

- a superkey for which no proper subset is a superkey: a superkey that is minimal
- Can be more than one for a relation

Primary Key

- a candidate key chosen to be the main key
- One for each relation,
indicated by underlining the key attributes

- Student(studno, name, ...)

▶ 1:10:17



Topic: keys Concept



SUPER KEY

minimal Candidate key [Let's Assume 6 Candidate key]

1 Select as → **Primary key**

Alternative/secondary key

▶ 1:14:08



Topic: Candidate Key

Key / Prime Attribute : The Set of Attribute which Present / belongs to Any (some) Candidate key.

Non key / Non Prime Attribute : The Set of Attribute which Not Present / Not belongs to Any Candidate key.

▶ 1:27:21



Topic: Finding Multiple Candidate Key

Procedure:

First find Any One Candidate key. that Attribute is Called key / Prime Attribute.

If $X_{\text{Attribute}} \rightarrow [\text{Prime Attribute}]$

▶ 1:36:26

Q R(ABCDEF) [A → BCDE, BC → ADE, E → F]

Solⁿ (A)⁺ = [ABCDEF]

A is Candidate Key — ①

Key/Prime Attribute = [A, B, C]

A → B already taken
A → C

If X Attribute → [Prime Attribute]

B > BC → ADE
C > (BC)⁺ = [BCADE]

(B)⁺ = [CB]

(C)⁺ = [C]

BC is Candidate Key — ②

2 C.K
A, BC (Any)

▶ 1:46:50



Keys Concept

[NAT]

#Q. R(ABCDEF) {A → B, B → C, D → CEF}

Find candidate keys for the relation R?

(A)⁺ = [ABC]

Key/Prime Attribute = [A, D]

(AD)⁺ = [ABCDEF]

If X Attribute → [Prime Attribute]

AD is Super Key

No multiple C.K

(A)⁺ = [ABC]

(D)⁺ = [DCEF]

C.K AD Any

AD is Candidate Key — ①

▶ 2:01:10

[NAT]



#Q. R(ABCD) { $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ }
Find candidate keys for the relation R?

key/
Prime
Attribute

$= (\check{A}, \check{D}, \check{C}, \check{B})$

$(A)^+ = (ABCD)$

A is Candidate key — (1)

If $X \xrightarrow{\text{Attribute}} \text{[Prime Attribute]}$

$D \rightarrow A$

$(D)^+ = (DAB C)$

D is C.K — (2)

$C \rightarrow D$

$(C)^+ = (CDAB)$

C is C.K — (3)

$B \rightarrow C$

$(B)^+ = (BCDA)$

B is C.K — (4)

u.c.k

A
B
C
D

Ans

▷ 2:07:43